Swaminathan Sundar

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Summary

- Ph.D. candidate in Chemical Engineering at Texas A&M University specializing in multiscale modeling, mathematical optimization, and data-driven decision-making for energy and manufacturing systems.
- 5+ years of experience with Python-based modeling and analytics (Pyomo, Pandas, NumPy, PyTorch), linear, nonlinear and mixed-integer programming (LP/NLP/MILP), stochastic optimization, and model predictive control (MPC/NMPC).
- First-author publications in *Computers & Chemical Engineering* applying optimization, TEA/LCA, and digital twins to large-scale planning and operations problems (e.g., DOE funded CCUS bio-refinery).

Technical Skills

Programming: Python (Pyomo, Gurobi, Pandas, NumPy, PyTorch, Keras, TensorFlow, scikit-learn), MATLAB, C++, Git, Linux/SLURM
Operations Research & Optimization: Stochastic & robust optimization (two-stage stochastic programming), linear, nonlinear and mixed-integer programming (LP/NLP/MILP), decision-making under uncertainty, supply chain & scenario/risk analysis, heuristic optimization, graph theory, dynamic and multiscale high-fidelity modeling, model predictive control (MPC/NMPC), time series based forecasting, surrogate modeling, global sensitivity analysis (Sobol), model-based design of experiments (MBDoE).
Modeling, Simulation & Solvers: Pyomo, Gurobipy, GAMS, MATLAB, LINGO, AMPL, Pyomo.DAE, gPROMS, Aspen Plus, COMSOL, Microsoft Excel, Gurobi, IPOPT, CPLEX, BARON, ANTIGONE, GLPK.
Analytics & Application Development: Streamlit, data wrangling and visualization in Python, techno-economic analysis (TEA), static and dynamic LCA (OpenLCA, GREET), and supply-chain modeling

Experience

Texas A&M Energy Institute, College Station, TX
Graduate Research Assistant **Aug 2021 - Present**

- Designed a generalized multiscale digital-twin framework linking unit operations, plant-level operations, and supply-chain decisions for emerging process technologies, demonstrated on algae-based CO_2 utilization and designed to be transferable to energy, chemical, and manufacturing networks.
- Built high-fidelity photobioreactor digital twins in Python/Pyomo with transport, reaction, & biological kinetics, coupled with global parameter estimation, Sobol sensitivity analysis, & structural identifiability analysis; developed a model-based design of experiments (MBDoE) framework using D_s-optimal designs on non-identifiable parameter subsets, reducing the parameter confidence region $\sim 24\times$ versus a full factorial design at half the experimental evaluations. Deployed as multi-stage pipeline on the TAMU HPRC cluster.
- Developed forecast-aware nonlinear model predictive control (NMPC) formulations embedding exogenous forecasts (e.g., solar irradiance) into closed-loop optimization, combining dynamic models and time-series data for real-time operational decision making.
- Developed a generalized techno-economic analysis (TEA) and life-cycle assessment (LCA) framework for supply-chain design and operations using a state–task network representation, automatically formulating mixed-integer optimization models to evaluate technology selection, capacity expansion, inventory management, and logistics tradeoffs against cost and emissions KPIs.
- Extended this framework to a two-stage stochastic optimization model that separates long-term investment decisions from scenario-dependent operations, enabling risk-aware planning under uncertain prices and demand profiles.
- Led modeling efforts for the DOE-funded Continuous Algae-based Carbon Capture and Utilization (CACCU) and NSF-funded CURB Engineering Research Center projects, delivering the mechanistic models and model-based analyses used across the program.

Carnegie Mellon University, Pittsburgh, PA
Graduate Research Assistant **Aug 2019 - Dec 2020**

- Modeled and optimized a photo-catalytic CO_2 utilization process in Pyomo, integrating Langmuir–Hinshelwood kinetics, annular fluid dynamics, and radiation models.
- Achieved $25\times$ higher CO_2 conversion by optimizing reactor length and recycle configuration; performed sensitivity analysis to identify key levers for design and operation.

Reliance Industries Limited, Navi Mumbai, India
Intern, Process Engineering (Reliance Project Management Group) **Jan 2018 - Apr 2018**

- Built Aspen Plus (Peng–Robinson) models of an industrial naphtha splitter (340 t/h feed, $\sim$55 components) as the baseline for evaluating a dividing wall column; used shortcut (DSTWU) analysis to select the reflux ratio and set the condenser pressure so the overhead could be condensed with cooling water at 45 °C.
- Validated a rigorous RadFrac model (40 stages) against plant datasheet data, matching product compositions to within about 4 percentage points (IC_5 55% vs. 51%, nC_5 25% vs. 29%) and CS_2 in the bottoms ($\sim$2 ppm).
- Wrote a technical report on distillation sequencing, thermal coupling, and dividing wall column configurations, covering their energy and capital trade-offs.

Selected Projects

Fleet Decarbonization Planning, Shell.ai Hackathon 2024 (Top 20 globally; team of four)

- Co-developed a MILP in Pyomo/Gurobi for a 16-year fleet transition problem (diesel, LNG, and battery-electric trucks), minimizing total cost of ownership under yearly carbon caps, demand by size and distance class, 10-year vehicle lifetimes, and a 20% annual sales limit.
- Built the Streamlit dashboard for running and comparing three fuel-price scenarios (median, lower, and upper bounds).

NLP for Review Classification with Transformers, Texas A&M University, Aug 2023 – Dec 2023

- Built an NLP pipeline on 174,000 Yelp reviews using Python (NLTK, PyTorch).
- Implemented custom positional encoding and a Transformer-based classifier; performed hyperparameter tuning and achieved 83% validation and 82.9% test accuracy.

Metaheuristic Optimization for Capacitated Vehicle Routing (CVRP), Texas A&M University, Jan 2022 – May 2022

- Designed a metaheuristic (2-opt and λ-interchange with Iterated Local Search) for CVRP to generate near-optimal routes.
- Benchmarked performance across problem sizes and analyzed scalability and runtime behavior.

Optimization and Quantum Annealing for Particle Track Reconstruction, Carnegie Mellon University, Aug 2020 – Oct 2020

- Formulated particle-track reconstruction from LHC data as both a mixed-integer optimization problem (Pyomo) and a QUBO model.
- Implemented simulated and quantum annealing-based solvers; evaluated solution quality and scaling with number of particles.

Education

Texas A&M University College Station, TX
Doctor of Philosophy in Chemical Engineering; GPA: 3.7/4.0 *Aug 2021 – Dec 2026 (expected)*
Advisor: Prof. Efstratios N. Pistikopoulos
Dissertation: *Multiscale Modeling and Optimization Framework for Digital Twin Development Applications*
Relevant Coursework: *Nonlinear Programming, Advanced Process Optimization, Data Mining and Analysis, Machine Learning, Deep Learning, Heuristics Optimization*

Carnegie Mellon University Pittsburgh, PA
Master of Science in Chemical Engineering; GPA: 3.7/4.0 *Aug 2019 – Dec 2020*
Advisors: Profs. Debangsu Bhattacharyya and Chrysanthos Gounaris
Relevant Coursework: *Advanced Process Systems Engineering, Product and Supply Chain Optimization, Quantum Integer Programming, Data Science in Chemical Engineering*

SRM Institute of Science and Technology Chennai, India
Bachelor of Technology in Chemical Engineering; CGPA: 9.07/10 *Aug 2014 - May 2018*

Teaching, Leadership, and Honors

- Teaching Assistant: Process Dynamics & Control; Fluid Operations in Chemical Engineering; Advanced Process Optimization.
- Peer reviewer, *Computers & Chemical Engineering*, *Bioresource Technology* and *Biotechnology Advances*.
- Contributed to multiple grant proposals, including the successful $26 million NSF CURB Engineering Research Center initiative.
- Vice President, Chemical Engineering Master's Student Association (ChEMSA), CMU: organized the first ChEMSA Research Symposium; supported Master's students during COVID-19.
- Business Development Executive, AIESEC Navi Mumbai: co-organized Youth Speak Forum 2018; helped raise INR 200k for the chapter.

Publications

1. Swaminathan Sundar, Rahul Kakodkar, and Efstratios N Pistikopoulos. Techno-economic analysis and life cycle assessment of a novel algae-based ccus technology. *Computers & Chemical Engineering*, page 109409, 2025
2. Rahul Kakodkar, Swaminathan Sundar, and Efstratios N Pistikopoulos. Hydrogen-based dense energy carriers in energy transition solutions. In *Handbook of Smart Energy Systems*, pages 2987–3007. Springer, 2023
3. Swaminathan Sundar, Vincent Xu, Bin Long, Susie Dai, Joshua Yuan, Yinjie J. Tang, and Efstratios N. Pistikopoulos. From dynamic modeling to optimal experiments: An mbdoe framework for scale-up of algae based ccus process. In preparation, 2026
4. Swaminathan Sundar, Vincent Xu, Dustin Kenefake, Yinjie J. Tang, and Efstratios N. Pistikopoulos. Forecast-aware nonlinear model predictive control of photobioreactors under solar irradiance uncertainty. In preparation, 2026